

Section 7.4 Value-added tax (VAT) – Discount

Discount

A 15% discount means the price is reduced by 15%.

If you get a 15% discount on the price of an item, you now pay only 85% of the price.

Example 1

- (i) A store offers a discount of 20% on all items during a sale.
Find the sale price of a television set which had a marked price of €780.
- (ii) A music shop offered a discount of €4.80 on DVDs with a marked price of €24.
Express this discount as a percentage.

- (i) If the discount is 20%, the sale price is 80% of the marked price.

$$80\% = 0.8$$

$$€780 \times 0.8 = €624$$

∴ the sale price is €624.

- (ii) The percentage discount is

$$\begin{aligned} \text{Percentage discount} &= \frac{€4.80}{€24} \times \frac{100\%}{1} \\ &= \frac{4.80 \times 100}{24} = 20\% \end{aligned}$$

∴ the percentage discount is 20%.

Value-added tax (VAT)

Value-added tax or **VAT** is a government tax that is added to many of the things that we buy. Most receipts for goods bought show how much VAT has been paid on the item.

The shop keeper has set his cash register to calculate the VAT and to add it on to the price of the item to give the total to be paid.

If an article is priced at €280 and the VAT rate is 23%, the price *including* VAT can be worked out as follows:

Using a calculator,

since $100\% + 23\% = 123\%$, we need to find 123% **of** the original amount.

$$123\% \times €280 = 1.23 \times €280 = €344.40$$

∴ the full price (the price including VAT) = €344.40.

Example 2

The rate of VAT on electrical goods is 20%.

- (i) Find the selling price of a washing machine priced at €650 + VAT.
- (ii) If the selling price of a laptop, including VAT, is €576, find the price of the laptop before VAT is added on.

- (i) VAT at 20% to be added on means 120% of the original price

$$\therefore 120\% \text{ of } €650 = 1.20 \times €650 = €780$$

- (ii) €576 represents 100% + 20%, i.e. 120% of the price before VAT is added on.

$$\therefore 120\% = 576$$

$$1\% = \frac{576}{120} = €4.80$$

$$100\% = 4.8 \times 100 = €480$$

$\therefore$ the price before VAT is added is €480.

Exercise 7.4

1. Without using a calculator, find the sale price of each of these articles:

Article	Marked price	Percentage discount
DVD recorder	€800	15%
Electric iron	€40	20%
Bicycle	€780	25%
TV set	€950	$12\frac{1}{2}\%$

2. Find the sale price for each of these marked prices:

- (i) Jumper: €45
- (ii) Boots: €80

SALE
Everything must go
30% off

3. The price of a certain car is €27 500.
During the month of December, the garage offers a discount of 15%.
What is the price of the car during December?



4. A man was given a discount of 20% when he bought a suit.
If he paid €256 for the suit, find the price of the suit before the discount by completing the following:
 80% of the price = €256
 1% of the price =
 100% of the price =
 Price before discount =
5. A woman paid €200 for a coat after a discount of 20% was given.
Find the price of the coat before the discount was allowed.
6. Find the selling price of each of the following:
 - (i) a television set marked €800 + VAT at 20%
 - (ii) a bicycle marked €650 + VAT at 20%
 - (iii) a games console marked €420 + VAT at 15%
 - (iv) a lawn-mower marked €760 + VAT at 10%.
7. Find the cost of hiring a car for 8 days at €75 per day plus VAT at 15%.
8. Find the selling price of a suite of furniture marked €850 + VAT at 20%.
9. A dinner in a restaurant for 4 people cost €148.
Find the total bill when VAT at $12\frac{1}{2}\%$ was added on.
10. Without using a calculator, find the total cost, including VAT, of each item.



VAT is 21%

11. A shopkeeper bought 200 pens at €2.40 each + VAT at 20%.
Find the total cost of the pens.
12. A bill from a garage for a car service consisted of €240 for labour and €360 for parts. VAT, at the rate of 10% for labour and 20% for parts, was added to the bill.
Find the total cost of the car service.
13. A man sold a house for €420 000.
The auctioneer charged him fees of 1% of the sale price plus VAT at 21% on these fees.
Find the amount of the bill from the auctioneer.
14. The price of an article is €33.60 when VAT at 20% is included.
Find the price of the article before VAT is added on.

15. The price of a meal for 8 people in a restaurant is €264 after VAT at 10% is included. Find the price of the meal before VAT is added.
16. The price of an overhead projector, including VAT at 25%, is €750. Find the price of the overhead projector before VAT is added.
17. Angela bought a bicycle for €744. This price included VAT at 20%. Find the price of the bicycle before VAT was added on.

Investigation:

Jack went to a builder's providers and bought building materials for €3750 plus VAT at 20%.

The providers told him that they would give him a discount of 10%. Jack considered whether he should take the discount first and then pay VAT on the reduced bill

OR

Pay the VAT first and then get the discount on the total.
Which should he choose to get the cheapest price?